Engine Model(s): PT6A

72-30-04

GAS GENERATOR CASE - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC03-001	Oil, Engine Lubricating
PWC05-003	Cloth, Abrasive
PWC05-014	Cloth, Abrasive, Scotch Brit
PWC05-101	Cloth, Abrasive
PWC05-118	Insert, Threaded
PWC07-023	Coating, Aluminum and Zinc
PWC09-002	Compound, Locking and Retaining
PWC11-033	Cleaner, Aqueous
PWC15-013	Inhibitor, Corrosion

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name	Application
PWC30271-100	Spreader, Nut	Obsolete - replaced by PWC31771
PWC31771	Spreader, Nut	Alternate to PWC30271-100
PWC32483	Cutter	
PWC32484	Gage	

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4. Fixtures, Equipment and Supplier Tools

The fixtures, equipment and supplier tools listed below are used in the following procedures.

Name

Locking Tool (Keensert)

5. Removal/Installation

A. Removal of Gas Generator Assembly (Heavy Maintenance)

(1) Should it be necessary to do approved repairs within the gas generator case:

(a) If required, remove the engine from the airframe and install in the stand (Ref. 72-00-00, REMOVAL/INSTALLATION and SERVICING).

(b) Remove the power section from the gas generator assembly and install in the stand (Ref. 72-00-00, SERVICING).

(c) Remove the compressor turbine disk and blade assembly (Ref. 72-50-02, Removal/Installation).

(d) Remove the combustion chamber liner (Ref. 72-40-00, Removal/Installation).

(e) Remove the compressor turbine vane ring, shroud housing assembly and exit ducts (Heavy Maintenance only) (Ref. 72-50-01, Removal/Installation).

B. Installation of Gas Generator Assembly

(1) Assemble the gas generator assembly:

(a) Install the compressor turbine vane ring, shroud housing assembly and exit ducts (Ref. 72-50-01, Removal/Installation).

(b) Install the combustion chamber liner (Ref. 72-40-00, Removal/Installation).

(c) Install the compressor turbine disk and blade assembly (Ref. 72-50-02, Removal/Installation).

(d) Remove the power section from the stand, and install on the gas generator assembly (Ref. 72-00-00, SERVICING).

(e) Install the engine in the airframe (Ref. 72-00-00, REMOVAL/INSTALLATION AND SERVICING).

6. Cleaning/Painting

A. Cleaning of Gas Generator Case

(1) Wash salt, smog and agricultural chemical deposits from the gas generator case with clean water. Dry with clean, dry compressed air.

(2) Clean corrosion buildup with a non-metallic brush.

NOTE: On engines with the diffused aluminide conversion-coated gas generator case , corrosion products (sacrificial) may become visually apparent. Coloring will be

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reddish brown to rust, which may give the impression that premature attack has occurred on the base material. Light brushing with a soft bristle brush will show that apparent "rust" is superficial. Depending on the coating vendor, the conversion coating color may be green to gray and glaze-like in appearance or may be similar to burnished aluminized enamel. Diffused intermetallic coating should be matte gray to brown. Damaged coating should be touch-up repaired (Ref. Para. 8.(2)).

7. Inspection/Check

A. Inspection of Gas Generator Case

(1) External:

(a) Examine for general condition, including cracks, distortion, corrosion and evidence of overheating.

(b) Minor corrosion on exposed surface of gas generator case may be removed (Ref. Approved Repairs).

NOTE: If condition of gas generator case indicates further inspection is required, remove fuel manifold adapters (Ref. 73-10-05, Removal/Installation) and spark igniters (Ref. 74-20-00, Removal/Installation).

1 Machined surfaces (bosses and pads) with corrosion pitting less than 0.010 inch deep and less than 75 percent of the surface is acceptable without repair.

(c) On engines INSTALLED in the airframe, examine the engine mounts for attachment and condition (Ref. Aircraft Maintenance Manual).

(d) On engines NOT INSTALLED in the airframe, examine the condition of the mounting pads and threaded holes. Damage to the threaded holes may be repaired (Ref. Approved Repairs).

(2) Internal (Heavy Maintenance Only):

(a) Examine the shanknuts on the flanges in the area of the No. 2 bearing. Loose or damaged nuts shall be replaced (Ref. Approved Repairs).

(b) Reject engine if two adjacent diffuser exit ducts are missing or if more than two non-adjacent diffuser exit ducts are missing.

(c) An engine missing a maximum of two non-adjacent diffuser exit ducts may continue in service only if all other existing diffuser exit ducts are within allowable limits and engine performance remains acceptable. Remove debris and sharp edges.

(d) Inspect for minimum clearance of 0.020 inch at the following locations (Ref. Fig. 201) :

1 Between adjacent diffuser exit ducts.

2 Between diffuser exit ducts and adjacent straightening vanes.

3 Between diffuser exit ducts and inner wall of the gas generator case.

(e) Fretting wear on the outer face and edge adjacent to the next diffuser exit duct is acceptable. Blend repair any sharp edges by removal of excessively thinned metal.

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(f) Any cracks on the inner or outer surfaces which will not converge and allow metal to become detached are acceptable (Ref. Details A, B and D) provided they are stop drilled (Ref. Approved Repairs). For any cracks which could converge and allow metal to become detached (Ref. Detail C), refer to Para. 8.

NOTE: Cracking of diffuser pipes may be associated with vibrations. The source of vibration should be identified and corrective action taken to reduce it.

(g) Loss of material on a maximum of four diffuser exit ducts is acceptable provided:

1 The detached material is removed from engine.

2 The size of the detached material does not exceed 0.50 x 0.80 inches on the inner surface, 0.30 x 0.30 inches on the outer surface (Ref. Details A and B, Ref. Fig. 201).

3 The hole in the inner surface is blended to remove uneven edges .

4 Fretted areas are blended to remove excessively thinned material .

NOTE: If any of the preceding limits are exceeded, the diffuser exit ducts must be replaced (Ref. Para. 8.F.).

(h) Examine the straightening vanes for cracks and missing material. Up to seven vanes with one side broken off is acceptable provided no more than two are adjacent to one another and sharp material is blended to make sure no further detachment of material.

NOTE: If the engine is kept in service, do a borescope inspection of the hot section and combustion chamber at recommended intervals (Ref. 72-00-00, INSPECTION, Periodic Inspection).

(3) Inspect gas generator case for corrosion:

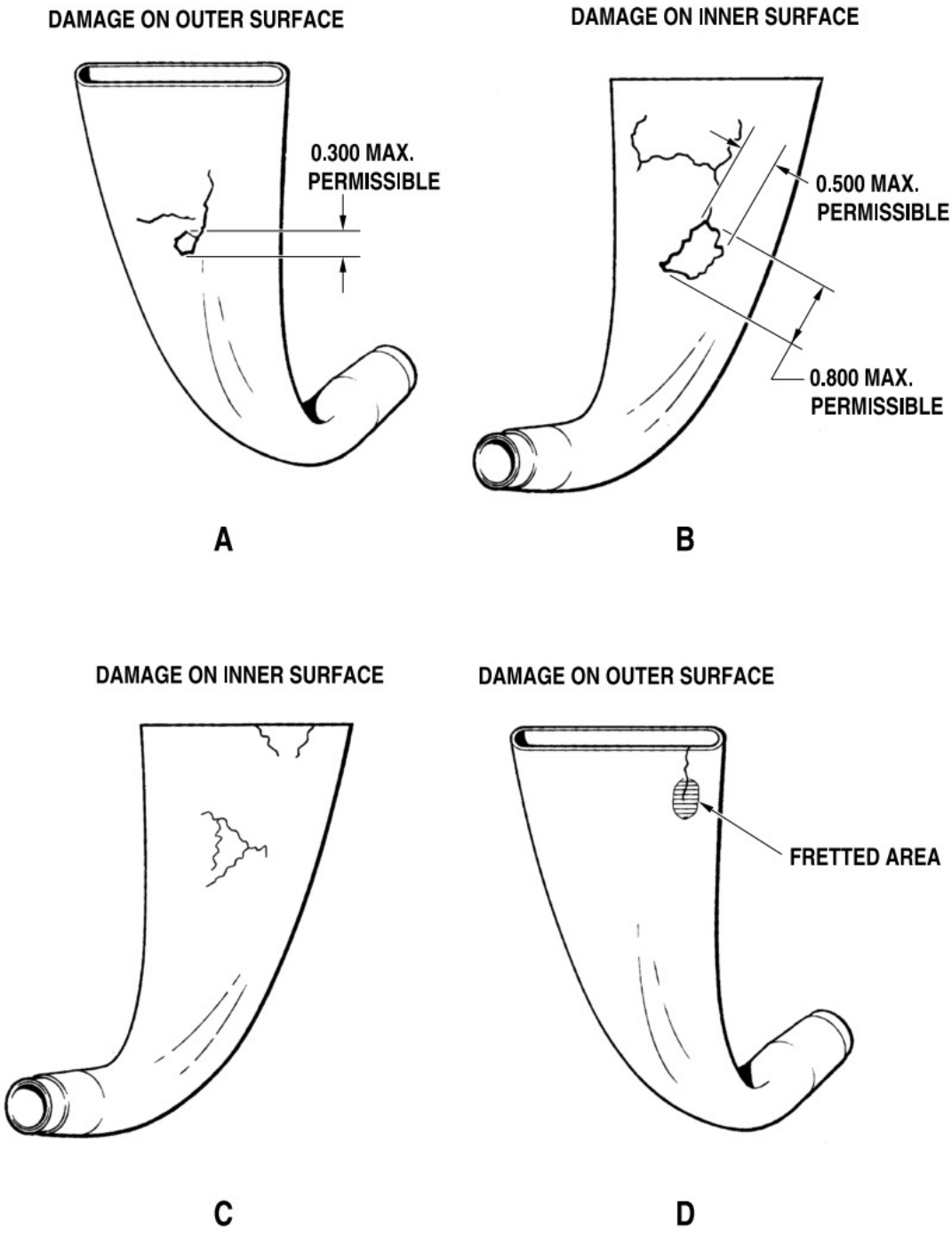
NOTE: For engines with a diffused aluminide conversion-coated gas generator case, the corrosion products may become visually apparent. Coloring will be reddish brown to rust, which may give the impression that premature attack has occurred on the base material. Light brushing with a soft bristle brush will show that apparent "rust" is superficial. Appearance of coating may vary. Damaged coating should be touch-up repaired (Ref. Approved Repairs).

(a) Minor corrosion on the exposed surfaces of the gas generator case may be removed (Ref. Para. 8.B.).

(b) If the condition of the corrosion exhibited on the exposed surfaces of the gas generator case indicates that further examination of the fuel manifold and igniter bosses is required, remove the fuel manifold adapters (Ref. 73-10-05, Removal/Installation) and spark igniters (Ref. 74-20-00, Removal/Installation). Examine the mounting pads, fuel nozzle bosses and machined surfaces for corrosion and wear. Isolated corrosion pitting not closely grouped, less than 0.010 inch deep, not covering more than 75 percent of the surface is acceptable without repair.

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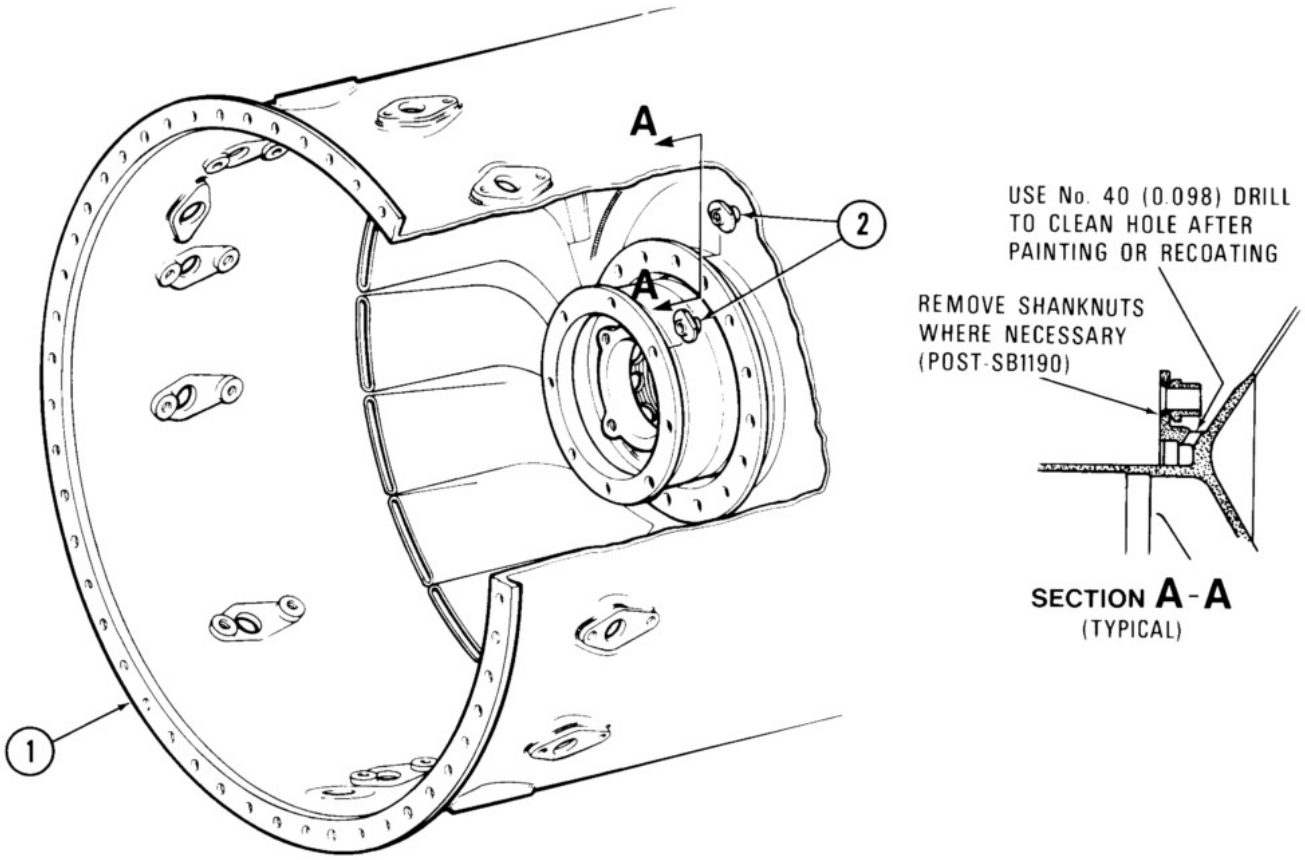
Figure 201 Diffuser Exit Ducts - Inspection



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Figure 202 Gas Generator Case - Repair



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1. Gas Generator Case
2. Shanknut

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8. Approved Repairs

A. Mount Pad Threads

(1) Drill hole 0.386 in. in diameter to a maximum depth of 0.490 in.

(2) Remove all swarf.

(3) Tap hole to 0.375 in. 24UNF-3B threads using plug and bottoming taps to a minimum depth of 0.400 in.

(4) Using inserting tool, install helicoil insert finishing 1 to 1½ thread pitch below surface.

(5) Remove driving tang of insert with break off tool.

B. Corrosion Removal

WARNING: GAS GENERATOR CASE MUST HAVE COOLED TO AMBIENT TEMPERATURE.

(1) Remove all visible corrosion with coarse abrasive cloth (PWC05-003) exposing bare metal. Use pad (PWC05-014) or a soft bristle brush to provide a clean surface.

NOTE: Do not use power tools for corrosion removal.

(2) Restore coating on gas generator case that exhibit small localized areas of coating loss :

(a) Clean all touch-up areas with cleaner (PWC11-033). If necessary, scrub area with a soft non-metallic bristle brush.

(b) Mask all exposed holes in fuel or igniter pads.

(c) Feather touch-up area with abrasive cloth (PWC05-101).

(d) Thoroughly agitate, on a paint shaker or equivalent, coating material.

(e) Apply to two or three brush, spray or damp sponge coats of (PWC07-023). Excess coating must be wiped off.

(f) Allow each coat of touch-up conversion coating to air dray for four hours between coats.

(3) Clean all touch-up areas with cleaner (PWC11-033). If necessary, scrub area with a soft non-metallic bristle brush.

(4) Mask all areas that are to remain free of corrosion inhibitor (PWC15-013).

(5) Apply inhibitor in accordance with manufacturer's instructions.

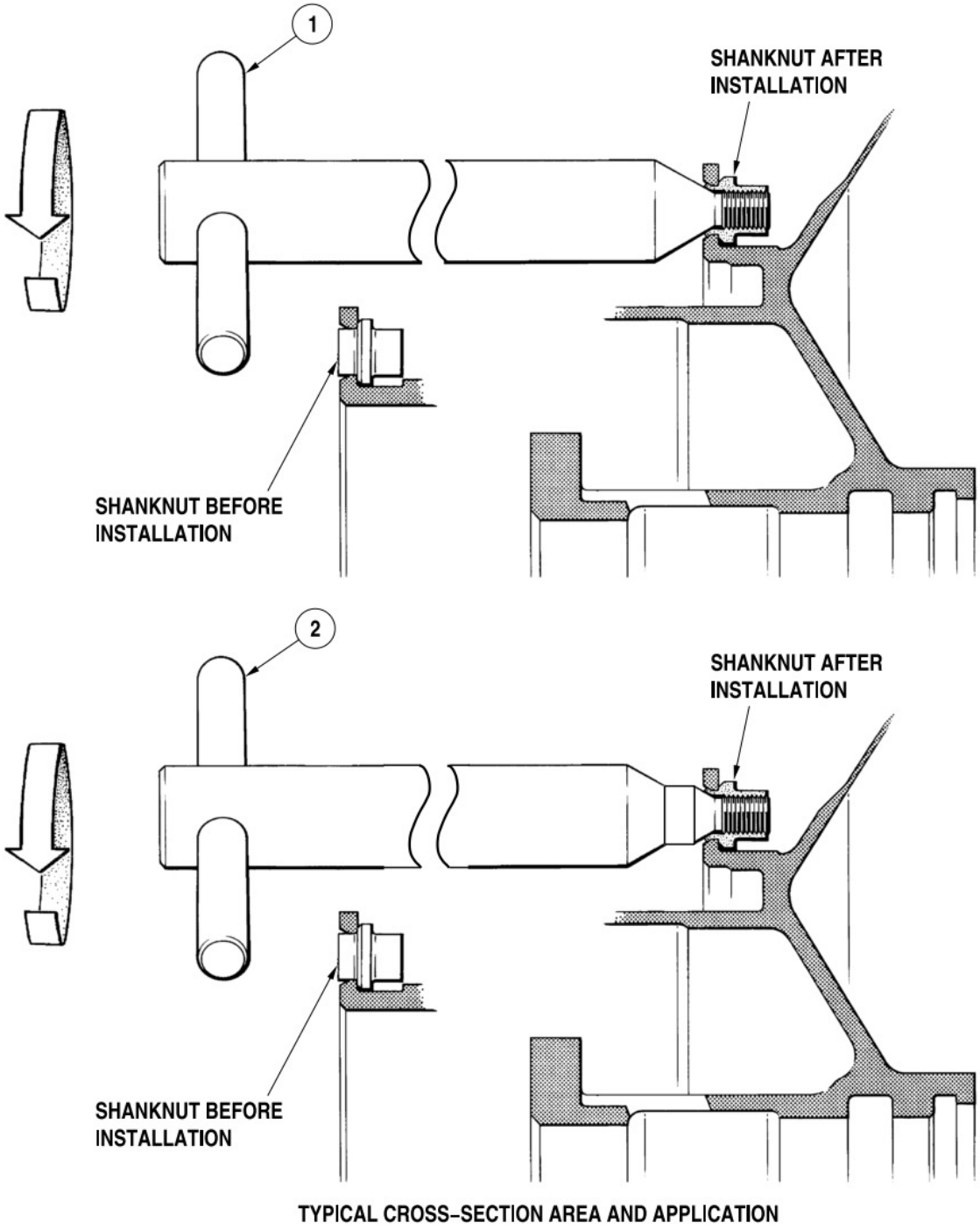
C. Repair of Igniter Boss Threads

(1) Repair damaged threads with a 0.5625-18 UNJF-3B tap.

D. Replacement of Shanknut (Heavy Maintenance Only) (Ref. Figs. 202 and 203)

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Figure 203 Shanknut Replacement



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1. Spreader (PWC30271-100)
2. Spreader (PWC31771)

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(1) Using a suitable drill, partially remove the flared end of shanknut (2, Fig. 202).

NOTE: Do not drill through to contact the flange.

(2) Remove the shanknut using a parallel pin punch at the drilled end to shear the weakened flare, and retrieve from the case.

(3) Install a new shanknut and hold against the flange (Ref. Fig. 203).

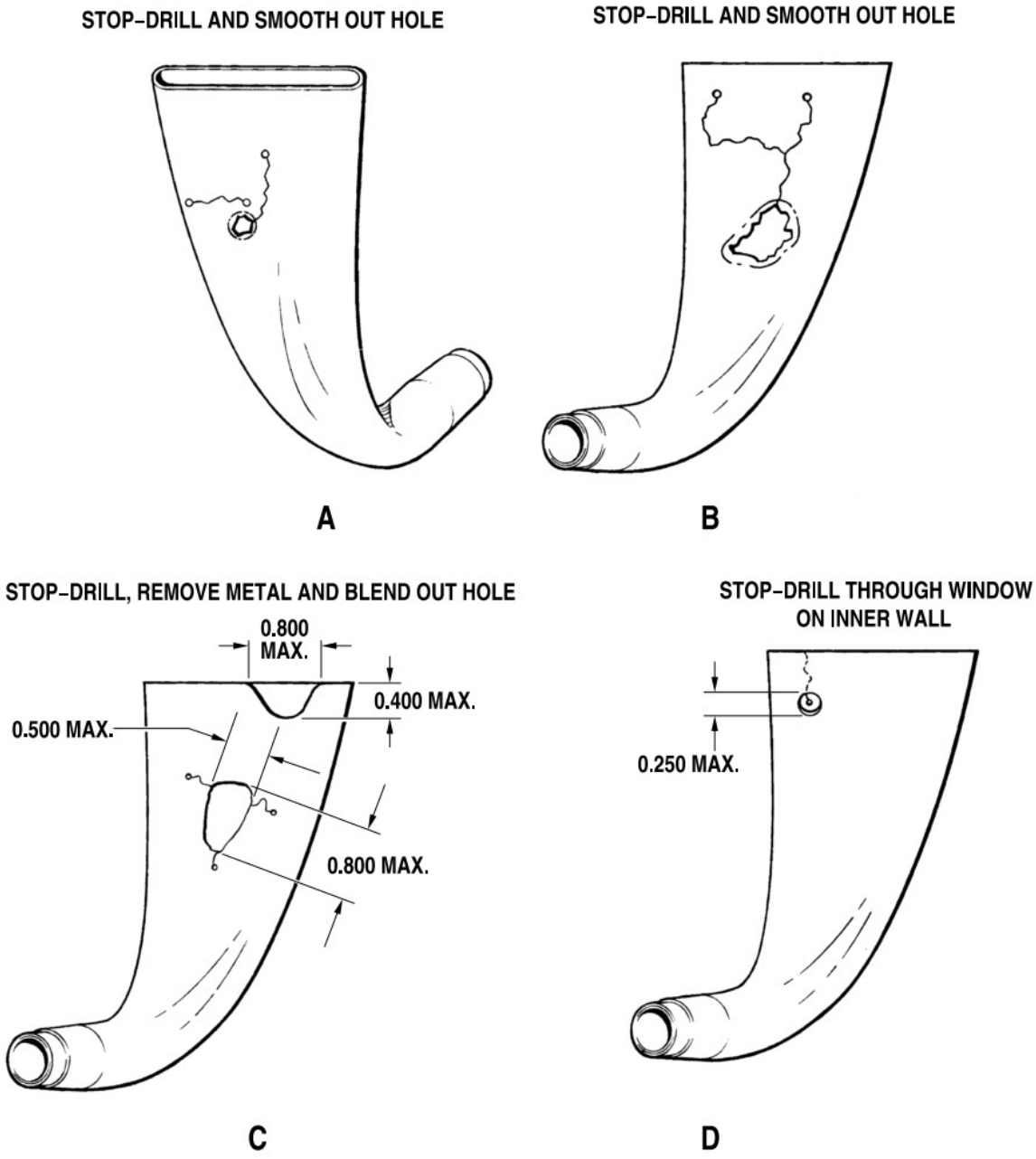
(4) Lightly lubricate the tapered portion of the shanknut spreader (1) (PWC30271-100) or (2) (PWC31771) with engine oil (PWC03-001) and screw into the shanknut until shank end is flared against the flange.

(5) Remove the spreader and examine the flared end of the shanknut for correct forming with no evidence of deformation or cracks.

E. Repair Diffuser Exit Ducts (Heavy Maintenance Only) (Ref. Fig. 204)

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Figure 204 Diffuser Exit Duct Repair



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(1) Stop drill the cracks using a 1/16 (0.0625) inch drill (Ref. Details A, B and C). Cracks in the outer wall of the diffuser exit ducts may be reached by drilling access holes in the inner walls using a 1/4 (0.250) inch drill, then stop drill the outer wall using a 1/16 (0.0625) inch drill (Ref. Detail D). Access holes are acceptable without repair.

(2) Remove the metal that can become detached if any converging cracks are allowed to progress.

(3) Blend out the areas where the metal has become detached, or has been removed (Ref. Details A, B and C), or where the metal has been thinned excessively due to fretting.

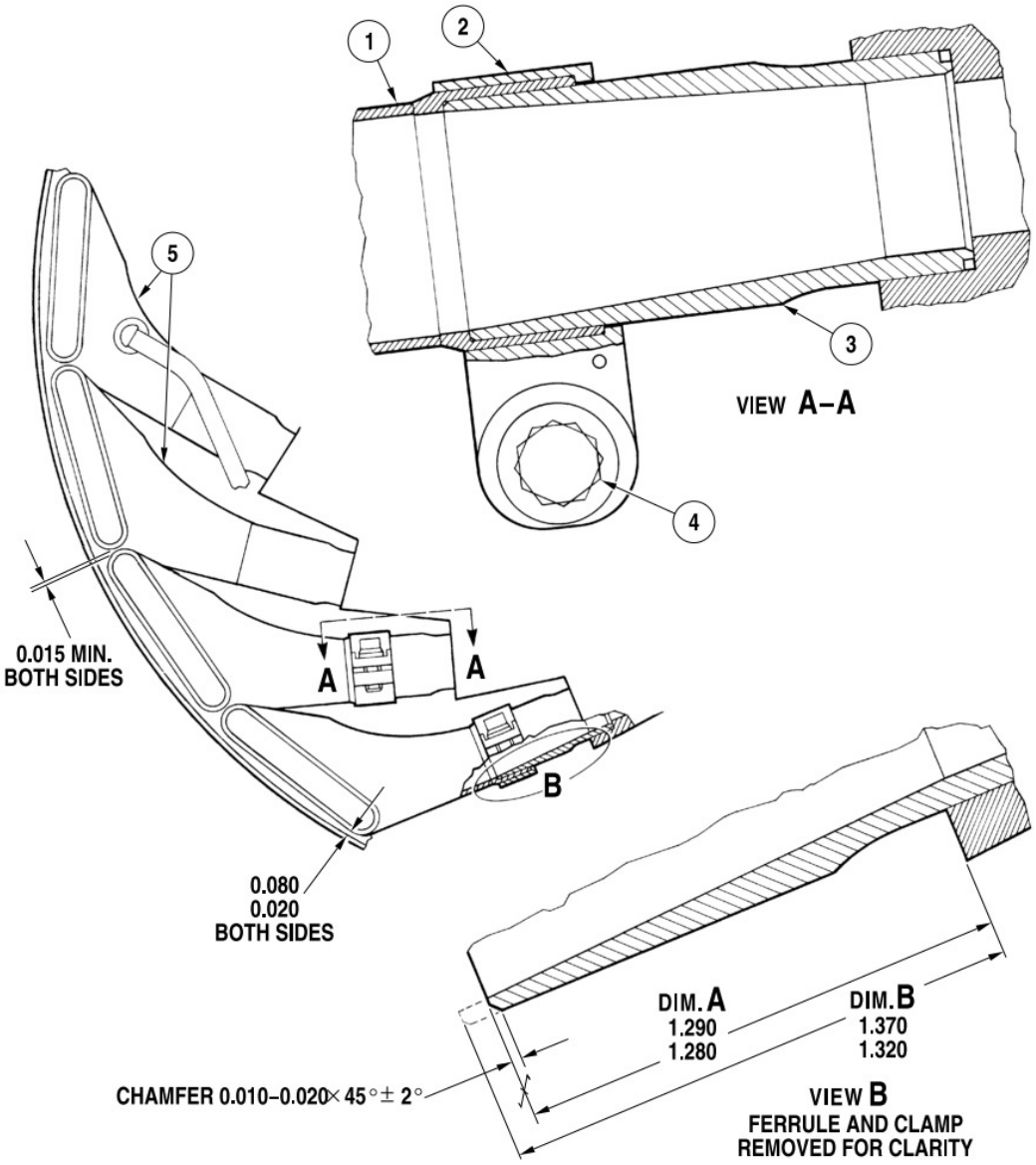
F. Replacement of Diffuser Exit Ducts

CAUTION: BEFORE MACHINING, MASK OFF THE HARDWARE IN THE VICINITY TO PREVENT THE INGRESS OF METAL PARTICLES INTO THE BEARING AREAS IN PARTICULAR AND THE ENGINE IN GENERAL.

(1) Replacement/repair of all diffuser exit ducts except at locations (5, Fig. 205):

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Figure 205 Diffuser Exit Duct Repair/Replacement



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1. Replacement Diffuser Exit Duct Assembly
2. Clamp
3. Stub
4. Bolt
5. Diffuser Exit duct Assembly (Repair not applicable at these locations)

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(a) Carefully remove the damaged diffuser exit duct(s) using the appropriate tooling; part-off to dimension (Ref. Dim. B, View B). To avoid damaging the gas generator case, insert a suitable shim (brass or steel) between the diffuser exit ducts and case wall during the part-off operation.

(b) Face (machine) the remaining stub to dimension 1.290 - 1.280 inch using a cutter (PWC32483) with a gage (PWC32484) and a suitable bar to apply pressure. Break sharp edges and chamfer as required (Ref. Dim. 'A', View B).

(c) Open up the clamp assembly (2) sufficiently to install onto the replacement diffuser exit duct assembly (1).

(d) Coat the OD of the stub (3) with sealing compound (PWC09-002) and install the assembly over the stub as shown.

(e) Tighten the bolt (4) progressively, maintaining the clearances shown, and finally tighten the bolt 60 to 65 lb.in. Fasten the bolt head with lockwire.

(f) On completion of repair, thoroughly clean and vacuum the gas generator case before and after removal of the masking tape. Remove the masking tape residue.

NOTE: Diffuser exit ducts (5) can only be replaced at an overhaul facility.

G. Repair Bleed Valve Mount Pad and/or Fuel Nozzle Pad Rear Threaded Holes

(1) Mask gas generator openings to prevent ingress of swarf.

(2) Open up the damaged hole(s) using a $^{9}/_{32}$ (0.281) inch drill. Chamfer 0.323 to 0.333 inch x 90 ± 5 degrees included angle.

(3) Tap the hole using 0.3125-18 UNC-2B tap.

(4) Install threaded insert (PWC05-118) fingertight as follows:

(a) Fuel nozzle inserts 0.010 to 0.015 inch above surface with keys aligned.

(b) Bleed valve inserts 0.010 to 0.015 inch below surface .

(c) Drive keys down with locking tool and hammer.

(5) Remove all masking material.

H. Replacement of Bleed Valve Mount and/or Fuel Nozzle Pad Damaged Threaded Inserts

(1) Remove insert material between keys and thread to $^{5}/_{32}$ (0.157) inch deep maximum with a $^{7}/_{32}$ (0.219) inch drill.

(2) Deflect keys inward, break off using suitable pin punch.

(3) Remove insert with suitable extraction tool.

(4) Install new insert (Ref. Para. G.).

I. Repair of Fuel Nozzle Pad Front Damaged Threaded Hole(s)

NOTE: This is a temporary (field level) repair. Holes repaired using these instructions will require removal of the threaded insert at overhaul.

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(1) Remove power section (Ref. 72-00-00, REMOVAL/INSTALLATION).

(2) Remove fuel manifold adapter(s) (Ref. 73-10-05, Removal/Installation).

(3) Remove combustion chamber liner (Ref. 73-10-05, Removal/Installation).

(4) Mask gas generator openings to prevent ingress of debris.

(5) Open up damaged front hole with a 0.203 (13/64) inch drill bit.

(6) Chamfer hole 0.220 - 0.250 inch diameter X 120 degrees ± 5 degrees.

(7) Tap hole and install helical insert (MS124655 or equivalent).

NOTE: The insert must be installed $^{3}/_{4}$ to one thread pitch below surface.

(8) Break and remove driving tangs.

(9) Remove all masking material.

(10) Clean inside gas generator case with suitable suction cleaner.

(11) Install combustion chamber liner (Ref. 73-10-05, Removal/Installation).

(12) Install fuel manifold adapter(s) (Ref. 73-10-05, Removal/Installation).

(13) Install power section (Ref. 72-00-00, REMOVAL/INSTALLATION).

J. Touch-up Protective Coating

WARNING: METHYL-ETHYL-KETONE IS HIGHLY INFLAMMABLE, OBSERVE PROPER PRECAUTIONS IN HANDLING. AVOID PROLONGED INHALATION OF FUMES FROM THIS SOLVENT.

(1) Clean grease and oil from touch-up areas with methyl-ethyl-ketone.

(2) Mask all exposed holes in fuel or igniter pad.

(3) Sand area(s) to bare metal with 180 grit aluminum-oxide abrasive cloth. Clean with dry, oil free shop air and wipe with a clean, dry cloth. Do not apply solvent to sanded surface and do not touch area with bare hands.

(4) Option 1: Apply SermeTel 249 (PWC07-023) and air dry 15-30 minutes or until dry to touch. Spray, brush or swab the coated surface with SermeTel 273 (PWC07-024) catalyst. Allow to stand one hour. Wash with water and dry with shop air.

(5) Option 2: Apply SermeTel 196 (PWC07-034) touch-up coating in two applications with either spray or brush. Air dry 30-60 minutes between coats. Air dry second coat 6-8 hours.

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